

A 2 Layer post-quantum Blockchain for Secure Electricity Trading in Smart Grids

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Abstract

The increasing complexity and demand of modern power systems have accelerated the transition toward decentralized smart grids, where prosumers can trade electricity efficiently and securely. While blockchain provides a trustless and monopoly-free platform for such transactions, existing solutions struggle with limited scalability, high latency, and significant vulnerabilities to emerging quantum attacks. To address these challenges, this proposal introduces a novel two-layer post-quantum blockchain architecture for secure and scalable energy trading.

Layer 1 adopts post-quantum cryptographic primitives to guarantee long-term security against quantum adversaries, while maintaining a lightweight consensus mechanism optimized for fast block generation in public networks. Layer 2 implements an off-chain payment and energy-routing mechanism inspired by payment-channel networks, enabling high-throughput, low-cost transactions with minimal interaction with the main chain. To ensure correctness in multi-hop transactions, a collateral-based security deposit system is designed for intermediary nodes, preventing misrouting and ensuring atomic settlement.

Furthermore, the proposed model integrates payment routing with physical energy routing—an essential requirement since electricity flows must follow physical grid constraints while financial settlements may traverse different logical paths. This work also addresses unresolved challenges such as multi-source energy provisioning, distributing energy across multiple physical paths, and recombining it at the destination. The research aims to formulate optimized routing algorithms under these constraints and evaluate system performance in terms of delay, throughput, scalability, and quantum resistance.

Overall, this proposal outlines the design, analysis, and validation of a secure, scalable, and quantum-resistant framework for decentralized electricity trading based on a two-layer blockchain architecture.

1 Introduction

The growing demand for electricity, along with the rapid expansion of distributed energy resources, has intensified the need for decentralized and efficient energy trading mechanisms. Modern smart grids integrate prosumers and diverse renewable sources such as rooftop photovoltaics and small-scale wind turbines, enabling peer-to-peer (P2P) energy exchange based on real-time supply and demand. This decentralization reduces dependency on centralized operators and fosters a more flexible, transparent, and competitive market [1, 2]. Despite these advantages, existing digital infrastructures struggle to support the high transaction volume required for large-scale energy trading. Public blockchains such as **Bitcoin** and **Ethereum** suffer from limited throughput—approximately **3 TPS** and **20 TPS**, respectively [3]—while realistic smart grid deployments demand **tens of thousands of transactions per second**.

In addition to scalability limitations, ensuring long-term security poses a major challenge. The emergence of quantum computing threatens classical cryptographic primitives, making **post-quantum cryptography (PQC)** essential for securing long-lived energy trading platforms. However, the computational and bandwidth overhead of PQC significantly increases on-chain latency, making real-time energy settlement on a purely Layer 1 architecture impractical.

To overcome these limitations, this research proposes a hybrid two-layer blockchain architecture. Layer 1 uses PQC-based digital signatures and key exchange mechanisms to provide quantum-resistant security, while Layer 2 employs high-throughput off-chain transaction frameworks such as payment channels, Rollups, and Plasma [4]. Most energy trading transactions are executed off-chain using a lightweight ledger mechanism, and only periodic aggregated commitments are stored on Layer 1. This drastically reduces congestion and enables low-latency, high-frequency trades suitable for smart grid environments.

A key innovation of the proposed system is the joint design of **energy routing** and **payment routing**. Since physical electricity flows must follow the constraints of the power network, whereas digital payments may traverse entirely different paths, separating them can lead to settlement inconsistencies, failed trades, or security vulnerabilities. By integrating both routing processes into a unified Layer 2 framework, the system ensures atomicity, consistency, and verifiable fairness in multi-hop and multi-source energy transfers.

This research aims to develop and evaluate a secure, scalable, and quantum-resistant framework for decentralized energy trading. It addresses key open challenges—including **security deposit mechanisms** for off-chain trust, **energy**

routing without direct cable connections, multi-path and multi-source energy aggregation, and **performance bottlenecks introduced by PQC**—to enable practical deployment of blockchain-based energy markets in next-generation smart grids.

2 Related Work

Blockchain technology has been extensively investigated in smart grid networks to enhance transparency, security, and operational efficiency in decentralized energy trading [5]. Despite these benefits, significant challenges remain, particularly regarding scalability and long-term security in the presence of quantum-capable adversaries. Prior studies have explored the integration of post-quantum cryptography (PQC) into smart grid infrastructures, focusing mainly on lattice-based constructions such as LWE¹ and RLWE² [6, 7]. Although consortium blockchains have been proposed as a potential solution [7], existing lightweight PQC algorithms still impose substantial computational and communication overhead. These delays, even when partially optimized, render purely on-chain PQC-based settlement impractical for the real-time requirements of energy markets.

Smart grid environments typically require transaction throughputs on the order of tens of thousands of transactions per second (TPS). In contrast, public blockchains such as Bitcoin and Ethereum are limited to approximately 3 and 20 TPS, respectively, whereas centralized payment networks like VisaNet routinely process more than 24,000 TPS [3, 8]. Although some consortium blockchain frameworks can exceed 1,000 TPS under controlled settings [9], real-world field deployments—such as the Walenstadt microgrid—have reported as little as 10 TPS [10]. Newer high-performance platforms, including Solana and Avalanche, demonstrate higher throughput (on the order of 2,000–3,000 TPS in practice) [11], yet still fall short of the dynamic, fine-grained transaction granularity required in large-scale P2P energy trading.

To overcome these limitations, Layer 2 scaling solutions have gained prominence. Approaches such as Plasma, Rollups, and the Lightning Network increase throughput by shifting most transactions off-chain and periodically committing aggregated results to the Layer 1 blockchain [12]. However, these mechanisms were originally designed for purely financial systems and are not directly applicable to physical asset trading, where the settlement process must be tightly coupled with physical constraints such as electricity flow paths, multi-source energy aggregation, and the temporal characteristics of grid operations.

This research therefore proposes a **post-quantum-secure blockchain augmented with Layer 2 energy trading channels**, integrating PQC for quantum-resistant security with high-throughput off-chain execution. The proposed architecture addresses fundamental challenges in decentralized energy markets, including (i) the design of **security deposit mechanisms** to ensure trustless multi-hop settlement, (ii) **routing of electricity in networks without direct cable connections**, and (iii) harmonization of digital payment routing with physical energy flow. The objective is to develop a secure, scalable, and practically deployable framework for next-generation smart grid energy trading.

3 Proposed Solution

This research introduces a hybrid blockchain framework with a post-quantum secure Layer 1 and a high-speed Layer 2 for efficient electricity trading in smart grids. It ensures **quantum-resistant security** and **low-latency transactions**. Layer 1 provides long-term resilience against quantum attacks, while Layer 2 employs off-chain payment channels to optimize efficiency. The framework incorporates **energy-backed security deposits** for trustless off-chain trading and an **electricity routing mechanism** to enable intermediary-based transactions without direct cable connections.

3.1 System Model and Assumptions

The proposed system enables decentralized energy trading among **producers**, **consumers**, and **intermediary nodes** when direct connections are unavailable.

Energy transactions occur via **bidirectional payment channels** using **multi-signature contracts** to ensure secure, trustless exchanges. High-frequency transactions are processed off-chain to minimize latency, while final settlements are recorded on the post-quantum Layer 1 blockchain for security and immutability.

Intermediary nodes facilitate energy routing, and all transactions are backed by **energy-backed tokens** to prevent fraud. By incorporating dynamic routing, liquidity checks, and path optimization, the system enhances scalability and efficiency in smart grid energy trading.

3.2 Energy-Backed Tokens

Energy-backed tokens represent digital assets tied to a specific quantity of electricity (e.g., 1 token = 1 kWh), ensuring a secure and verifiable means of energy exchange within decentralized smart grids. These tokens enable efficient, trustless transactions and facilitate intermediary-based energy trading when direct cable connections are unavailable.

¹Learning With Errors

²Ring Learning With Errors

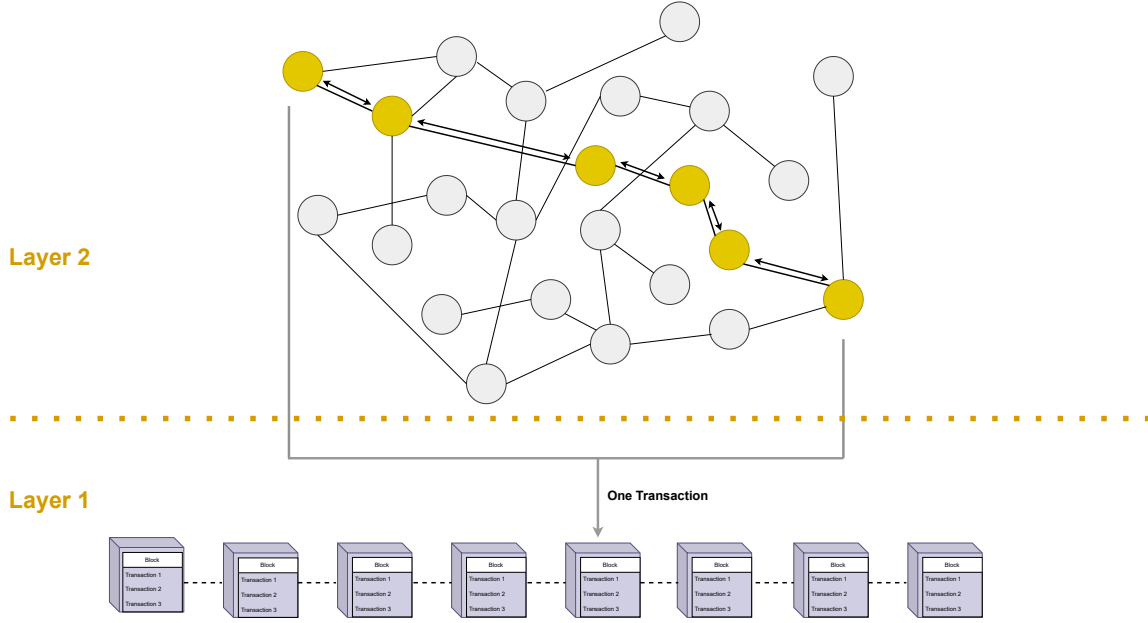


Figure 1: Overview of two layer transactions in smart grid network

Key Properties:

- **Energy Pegging:** Each token corresponds to a measurable unit of energy, ensuring intrinsic value.
- **Collateralization:** Tokens are issued only when an equivalent amount of electricity is committed to the network.
- **Redemption:** Consumers can redeem tokens for actual energy or trade them within the network.
- **Stable Valuation:** Since energy has inherent utility, token volatility remains lower than traditional cryptocurrencies.

Implementation in the Proposed System:

- **Minting Process:** Energy producers receive energy-backed tokens upon committing electricity, locked in a smart contract.
- **Transaction Mechanism:** Tokens are exchanged via off-chain payment channels, reducing latency and blockchain congestion.
- **Final Settlement:** Transactions are periodically recorded on the post-quantum Layer 1 blockchain to ensure security and immutability.
- **Security Deposit:** Tokens act as collateral, preventing fraudulent activities in off-chain trading.
- **Energy Routing Integration:** Tokens facilitate multi-hop transactions when direct energy transfer is unfeasible.

This tokenized approach enhances scalability, efficiency, and security in decentralized energy trading, bridging blockchain transactions with real-world electricity exchange.

3.2.1 Optimization Problem for Electricity Routing

To optimize electricity routing in decentralized smart grids, we formulate a **multi-objective optimization problem** using **Mixed Integer Linear Programming (MILP)**. The goal is to select the optimal energy transfer path that minimizes cost, delay, and congestion while ensuring liquidity.

Problem Definition: The smart grid network is modeled as a weighted directed graph $G(\mathcal{N}, \mathcal{E})$, where nodes $\mathcal{N}$ include **producers**, **consumers**, and **intermediaries**. Each energy transfer link $(i, j) \in \mathcal{E}$ has associated parameters: **transmission delay** (d_{ij}), **transaction fee** (c_{ij}), **congestion level** (l_{ij}), and **available energy balance** (b_{ij}).

Objective Function: The optimization minimizes the weighted sum of cost, delay, and congestion:

$$\min \sum_{(i,j) \in \mathcal{E}} (w_1 c_{ij} + w_2 d_{ij} + w_3 l_{ij}) x_{ij} \quad (1)$$

where x_{ij} is a binary decision variable indicating whether energy is routed through (i, j) .

Constraints:

- **Flow Conservation:** Ensures valid energy routing from source to destination.
- **Liquidity Constraint:** Requires that intermediary nodes have sufficient energy balance.
- **Path Selection:** Limits each node to participating in at most one routing path.
- **Energy Conservation:** Guarantees that transferred energy matches the requested amount.

Solution Approach: The problem can be solved using **Linear Programming (LP)**, **MILP**, or heuristic methods like **Ant Colony Optimization (ACO)** and **Genetic Algorithms (GA)** for real-time routing decisions. The optimization ensures cost-efficient, low-latency, and balanced electricity routing while maintaining liquidity, enabling scalable decentralized energy trading.

3.3 Final Settlement on the Layer 1 Blockchain

Upon channel closure, the final state of transactions is committed to the post-quantum Layer 1 blockchain. This guarantees transaction integrity, prevents double-spending, and provides an immutable record of energy trading activity. The integration of post-quantum cryptography at this level ensures long-term security, protecting against potential future quantum threats.

4 Advantages of the Proposed Model

- **Low Latency:** Off-chain transactions significantly reduce processing time.
- **Security:** Layer 1 ensures post-quantum security for final settlements.
- **Scalability:** Multiple energy transactions can be executed off-chain before settlement.
- **Energy Barter:** Intermediary routing enables energy trading even when direct connections are unavailable.

5 Expected Results

- 20–30× reduction in settlement delay compared to Layer 1-only systems,
- 50–100× scalability improvement via off-chain transactions,
- Guaranteed post-quantum security for long-term usage.

5.1 Implementation

For the proposed blockchain platform, **Hyperledger Fabric** was selected due to its modular architecture and flexibility, which allows customization of identity management, consensus, and cryptographic modules. Although Fabric is originally designed for permissioned networks, we adapted it to conceptually simulate a *public blockchain* environment.

5.1.1 Network Setup

A simulated network of 10 nodes was deployed in Mininet, including two independent organizations (**Org1** and **Org2**), each with two Peer nodes. A public channel **energychannel** was created for energy trading transactions, and smart contracts (Chaincode) were developed in Go to manage energy tokens, transaction logic, and settlement.

The ordering service used Solo for simplicity, though it can be upgraded to Raft. To emulate public blockchain behavior:

- Membership Service Provider (MSP) restrictions were relaxed, allowing nodes to join freely.
- Private channels were disabled; all transactions are stored on a shared ledger to ensure transparency.
- Consensus was adapted to simulate time-based competition for block creation.

Post-Quantum Cryptography with NTRU

The Fabric key-generation tool `cryptogen` was modified to replace classical algorithms (e.g., ECDSA) with the lattice-based post-quantum algorithm **NTRU**. Key changes include:

1. Replacing `ecdsa.GenerateKey()` with NTRU key generation using Go libraries (`pqcrypto`) or `liboqs`.
2. Designing a custom JSON-based identity format to replace X.509 certificates.
3. Updating MSP logic to validate NTRU public keys and signatures.
4. Ensuring peers and transaction validation use NTRU cryptography throughout the network.

5.1.2 Implementation Challenges

- NTRU keys are incompatible with standard X.509 certificates, requiring a custom identity format.
- MSP validation logic had to be rewritten for NTRU keys and signatures.
- Maintaining both classical and post-quantum setups for performance and security comparisons demanded modular design.

5.1.3 Network Configuration Summary

Table 1: Hyperledger Fabric Network Configuration

Component	Configuration
Organizations	2 (Org1, Org2)
Peer Nodes	2 per organization
Ordering Service	Solo (upgradeable to Raft)
Channel	Public channel <code>energychannel</code>
Smart Contracts	Manage energy token trading and settlements
Ledger Database	LevelDB
Chaincode Language	Go

Table 2: Comparison of Classical vs. NTRU Cryptography

Feature	Classical (ECDSA)	Post-Quantum (NTRU)
Public Key Format	X.509 certificate	Custom JSON/PEM format
MSP Compatibility	Full	Requires code modification
Key Generation Time	Short	Longer (depending on library)
Public Key Size	~256 bits	~1040 bits
Quantum Security	Vulnerable	Resistant
Tools Used	<code>cfssl</code> , <code>openssl</code>	<code>pqcrypto</code> / <code>liboqs</code>

Table 3: Key Technical Modifications for NTRU Support

Module	Changes Applied
cryptogen	Replaced ECDSA key generation with NTRU; updated identity structure
cfssl	Disabled X.509 certificate generation
MSP Handler	Added support for NTRU key loading and signature verification
Peer Config	Configured peers to use NTRU keys for authentication
Custom Identity Format	JSON structure with public key, node ID, and role

This implementation provides a controlled environment for simulating a public, post-quantum-resistant blockchain for smart grid networks, preserving Hyperledger Fabric’s flexibility and performance while enabling evaluation of NTRU cryptography.

5.2 Performance Comparison Metrics of the Proposed Network

The simulation results of the proposed blockchain framework for smart grids are evaluated against the SETS framework [13] across three main performance metrics:

- Total key distribution delay among network nodes.
- Computational time for cryptographic operations and signature verification.
- Block preparation and finalization time.

For each metric, five configurations are compared:

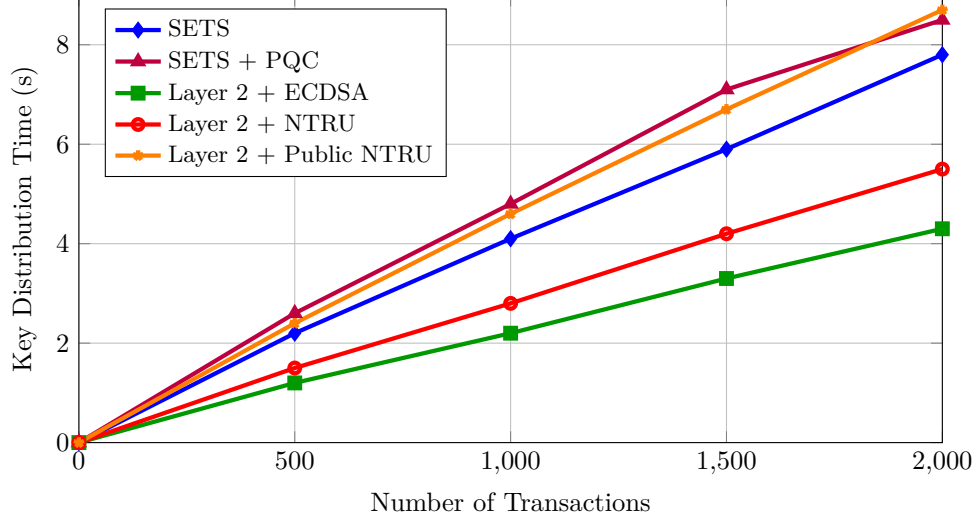
1. Layer 2 network with post-quantum cryptography (PQC) and private setup.
2. Layer 2 network with PQC and public setup.
3. Layer 2 network with classical cryptography (ECDSA) and public setup.
4. SETS framework with classical cryptography.
5. SETS framework with post-quantum cryptography (NTRU).

Comparing these five configurations highlights the impact of combining post-quantum cryptography with Layer 2 scaling on performance, scalability, and network security.

5.3 Comparison of Total Key Distribution Delay

One of the main challenges in blockchain-based energy trading is the delay in key distribution among nodes to initiate transactions. This delay can be caused by cryptographic computations, queuing in the main chain, or consensus mechanisms. Figure 2 shows the comparison across five different models:

Figure 2: Comparison of key distribution delay in five different network models

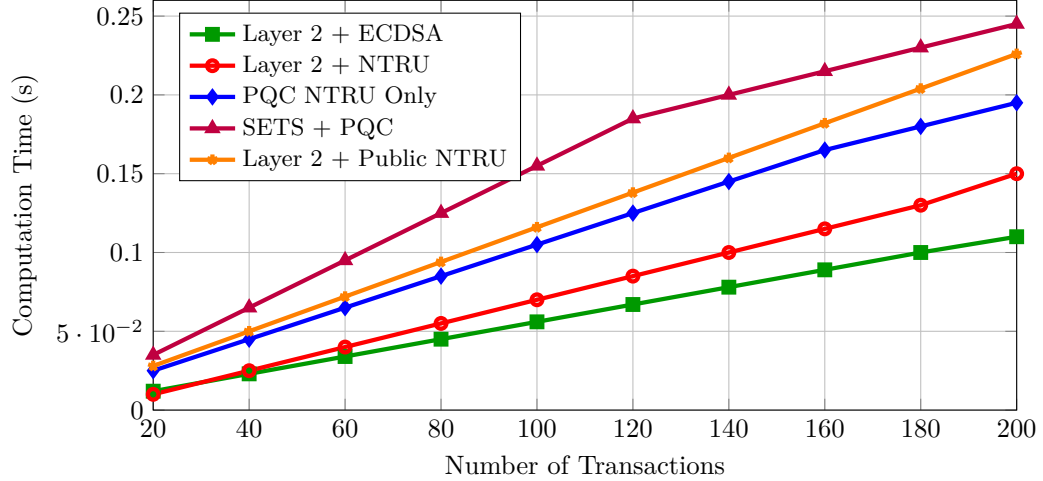


As observed in Figure 2, the Layer 2 architecture significantly reduces key distribution delay by offloading most transactions off-chain. Only 10% of transactions are processed on the main chain, while the rest are executed in Layer 2 payment channels. Using NTRU adds a small additional delay compared to ECDSA, but the post-quantum security benefits justify this increase. Public setups show slightly higher delays due to key management overhead and X.509 compatibility requirements.

5.4 Comparison of Cryptographic Computation and Verification Time

This section compares the computational time for cryptographic operations and signature verification across the five models. Figure 3 shows the results.

Figure 3: Comparison of cryptographic computation and verification time in five models



The lowest computation time corresponds to the Layer 2 ECDSA model due to lightweight cryptography. The Layer 2 NTRU model consumes slightly more time but is still significantly faster than fully on-chain models like SETS with PQC.

5.5 Comparison of Block Preparation Time

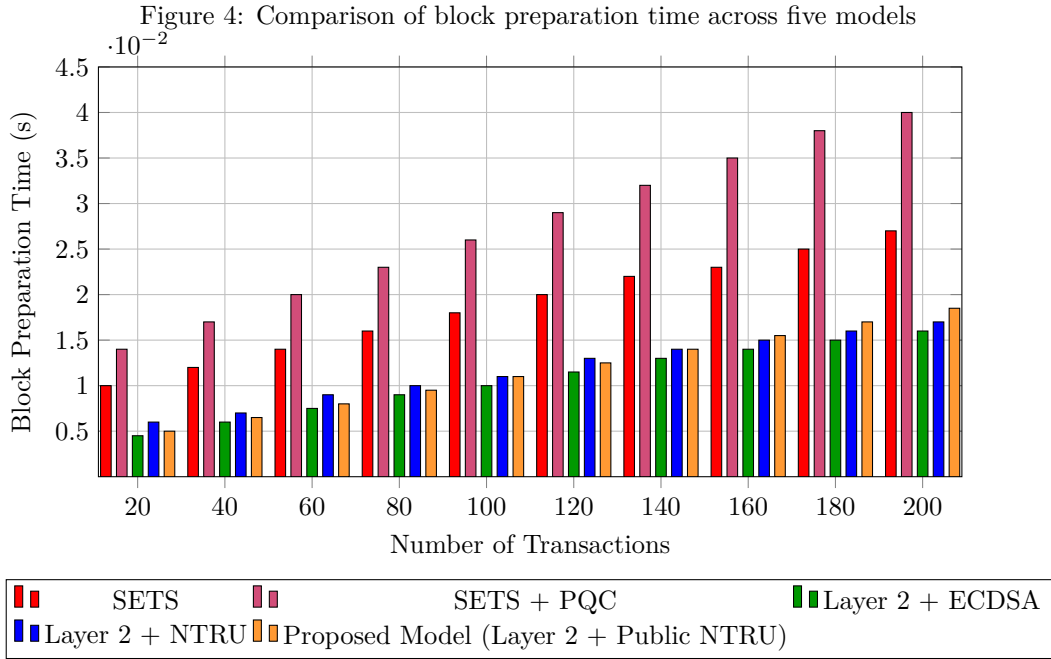


Figure 4 compares block preparation times. The proposed Layer 2 + Public NTRU model consistently outperforms other approaches. On average, block preparation time is reduced by approximately 30% compared to SETS and 10% compared to Layer 2 + NTRU. This improvement is due to:

- Offloading most transactions to Layer 2 channels, reducing on-chain processing.
- Efficient implementation of post-quantum cryptography (NTRU), which maintains high security while minimizing computation time.
- Optimized block structure storing only aggregated, hashed Layer 2 transactions, lowering block size and complexity.
- Layer synchronization and lightweight consensus mechanism for fast finalization on the public blockchain.

Overall, the combination of scalable Layer 2 solutions and post-quantum cryptography reduces block preparation time without compromising security or transparency.

6 Conclusion

This research proposes a Layer 2 blockchain algorithm to optimize energy trading in smart grids. By leveraging off-chain payment channels, it reduces transaction latency while ensuring final security via a post-quantum blockchain.

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